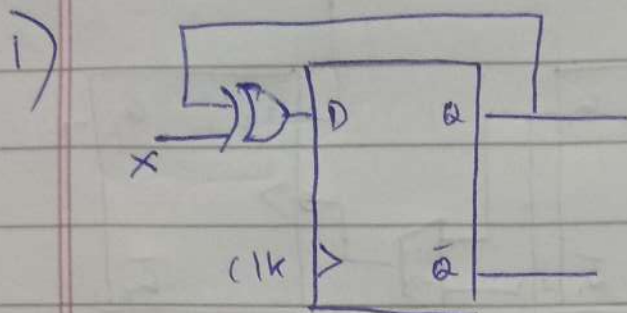


# DAY-4 Assignment

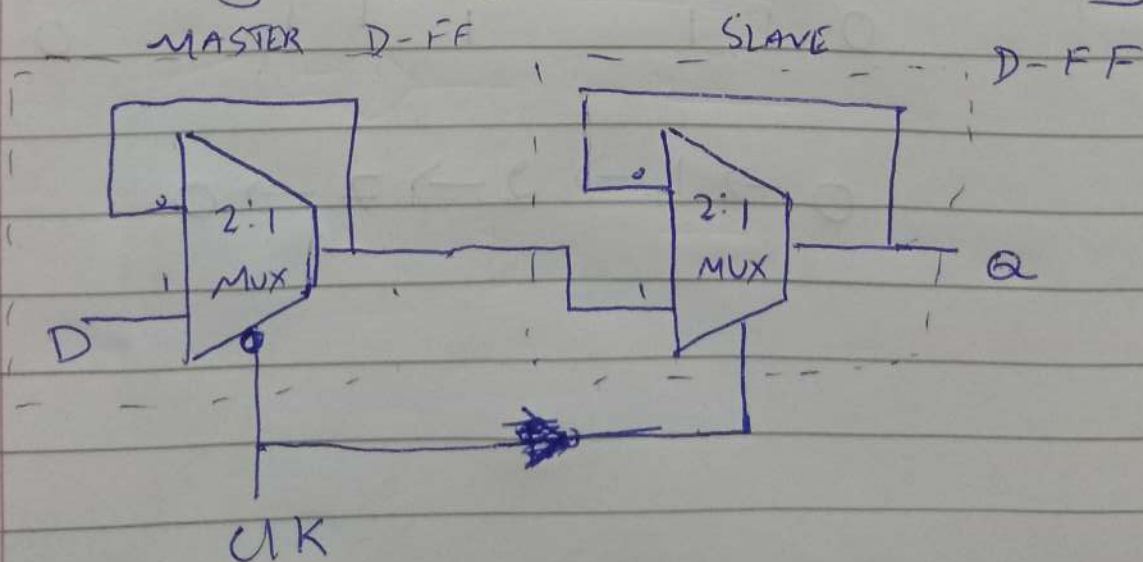


Next State Table

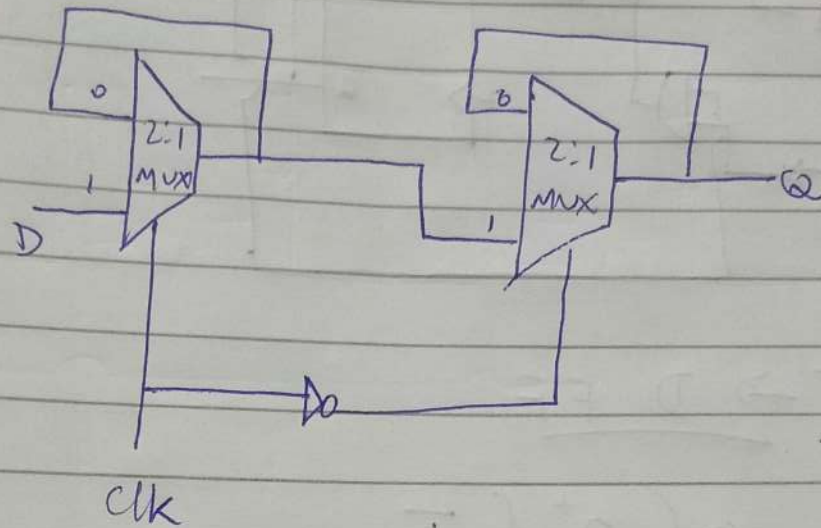
| X | Q | Q <sup>+</sup> |
|---|---|----------------|
| 0 | 0 | 0              |
| 0 | 1 | 1              |
| 1 | 0 | 1              |
| 1 | 1 | 0              |

The above table acts similar to T-FF  
 $\therefore$  T-FF

2) +<sup>ve</sup> edge triggered D-FF using 2:1 MUX



3) -ve edge triggered.



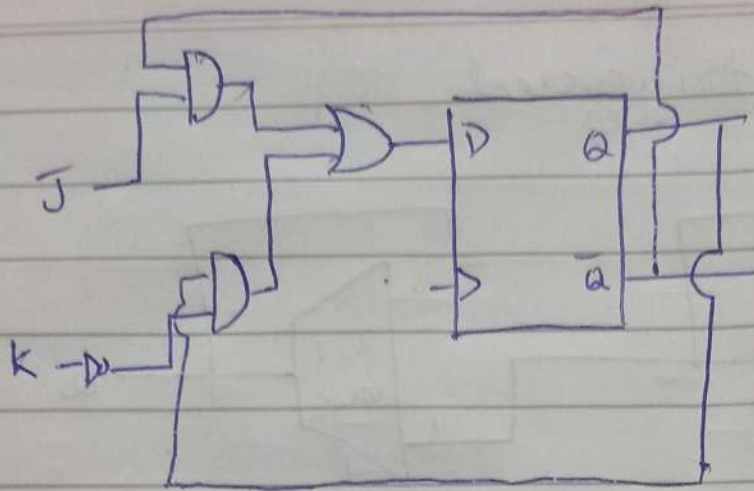
4)  
a) D FF  $\rightarrow$  JK

| J | K | $Q$ | $Q^+$ | D |
|---|---|-----|-------|---|
| 0 | 0 | 0   | 0     | 0 |
| 0 | 0 | 1   | 1     | 1 |
| 0 | 1 | 0   | 0     | 0 |
| 0 | 1 | 1   | 0     | 0 |
| 1 | 0 | 0   | 1     | 1 |
| 1 | 0 | 1   | 1     | 1 |
| 1 | 1 | 0   | 1     | 1 |
| 1 | 1 | 1   | 0     | 0 |

Using K-map -

$$D = Q\bar{K} + \bar{Q}J$$





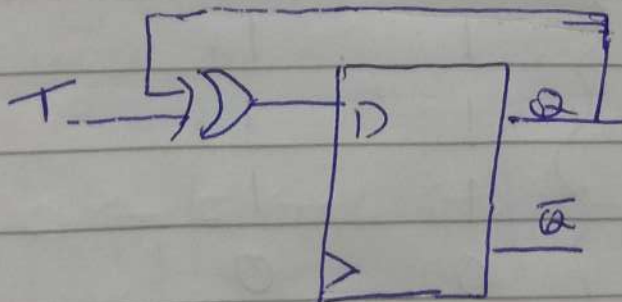
b)  $T \rightarrow \underline{D \text{ FF}}$

| D | Q | $Q^+$ | T |
|---|---|-------|---|
| 0 | 0 | 0     | 0 |
| 0 | 1 | 0     | 1 |
| 1 | 0 | 1     | 1 |
| 1 | 1 | 1     | 0 |

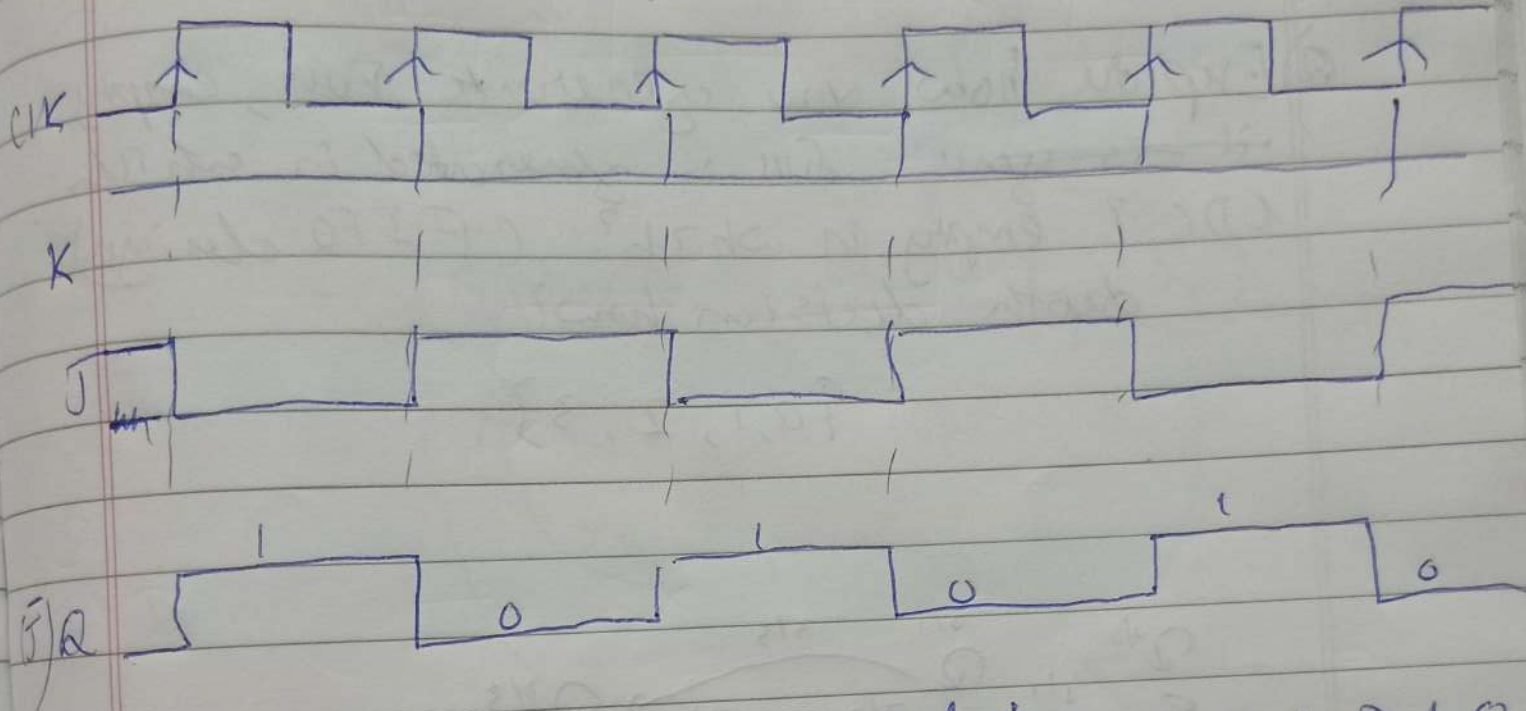
Using K-map

$$T = \bar{D}Q + D\bar{Q}$$

$$= D \oplus Q$$



5) Timing diag:  
(assuming  $Q=0, \bar{Q}=1$  initially)



$\therefore$  o/p sequence would be 1, 0, 1, 0, 1, 0